Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

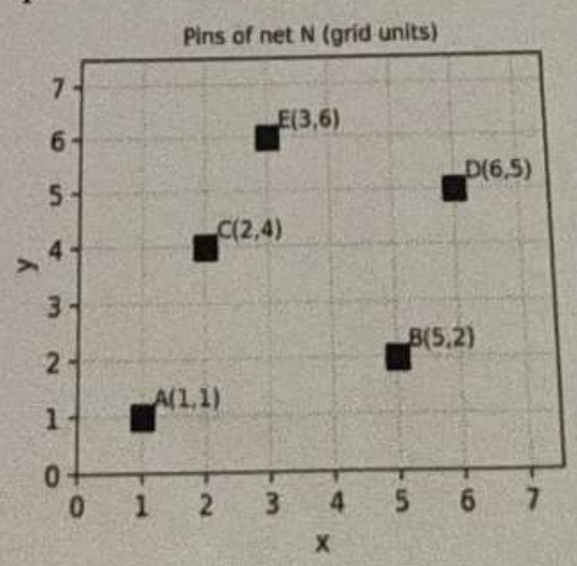
Continuous Assessment Test (CAT) – II September 2026

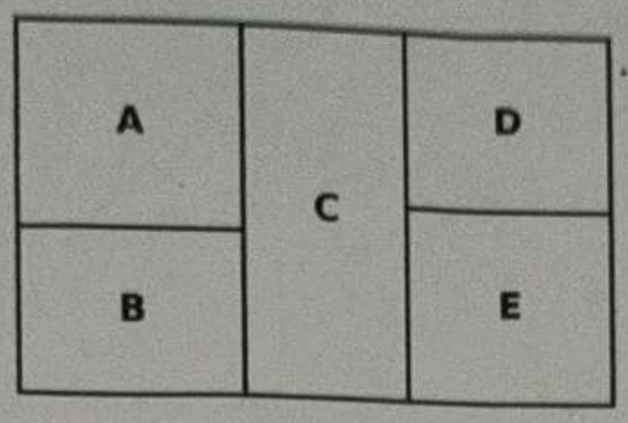
Programme	:	**B.Tech (VLSI Design and Technology)**	Semester	:	***Fall 2026-27***
Course Code & Course Title	:	**BEVD303L CAD for IC Design**	Class Number	:	***CH20262701019 82***
Faculty	:	**Dr. C. Sridhar**	Slot	:	***G1+TG1***
Duration	:	**90 Minutes**	Max. Mark		***50***

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		An SoC is being floorplanned. It contains five hardened IP blocks bought from a vendor: an SRAM macro, a PLL, a video decoder, a CPU core and an ADC. Each block is delivered with a fixed size and a fixed orientation, so no block may be rotated. Their dimensions (w × h, in mm) are listed below. **Block** | **1** | **2** | **3** | **4** | **5** **w × h** | 2 × 3 | 3 × 3 | 5 × 2 | 3 × 4 | 3 × 1 **(a)** For the Polish expression E = 12V3H45HV, draw the slicing tree and the corresponding floorplan. Determine the width, height, area and dead space of the floorplan. **(4 Marks)** **(b)** Apply the Wong–Liu move M1 by interchanging the neighbouring block numbers 3 and 4, and the move M2 by interchanging the last two operators of E. In each case write the new Polish expression and find the area of its floorplan. **(3 Marks)** **(c)** State whether each of the following is true or false, giving one sentence of justification in each case. **(3 Marks)** (i) Every slicing floorplan corresponds to exactly one slicing tree. (ii) A sequence pair can represent a non-slicing floorplan. (iii) Stockmeyer's algorithm finds the minimum-area orientation of the blocks of any floorplan in polynomial time.	10	3	K2
2		The figure 1 shows the pin locations of a five-pin net N, placed on a unit grid.  Figure 1 **(a)** Estimate the wirelength of net N using (i) the semi-perimeter wirelength, (ii) the complete graph model, and (iii) Streiner tree length **(6 Marks)**	10	4	K3

		(b) Differentiate between Quadratic placement and Force-directed placement. **(4 Marks)**			
3		An SoC architect has depicted in figure 2 has the relative arrangement of five hard IP blocks A–E.  Figure 2 **(a)** Identify the cut lines in the sketch. Draw the corresponding slicing tree and write the normalized Polish expression of this floorplan. **(3 Marks)** **(b)** Using Stockmeyer's algorithm, construct the shape function of every leaf and compose them bottom-up through your slicing tree. **(5 Marks)** **(c)** Determine the minimum area floorplan, trace back the orientation of every block, and state the dead space. **(2 Marks)**	10	3	K3

Block	A	B	C	D	E
w × h	2 × 4	3 × 2	2 × 5	3 × 3	1 × 3

4		Eight standard cells A–H of equal area are to be placed on a 2-row × 4-column array of unit slots by min-cut placement, using alternating cut directions: a vertical cut first, then a horizontal cut inside each half. All nets are two-terminal and carry unit weight. **(a)** The first (vertical) cut has to split the cells into two balanced groups of four. For each of the three candidate partitions {A, B, C, D}, {A, B, E, F} and {A, C, E, G}, list the nets that are cut and state the cut size. Which partition does min-cut placement choose, and which cells go to the left half of the die? **(3 Marks)** **(b)** Evaluate all three balanced splits, state the cut size of each, and give the minimum-cut split. Write down the resulting quadrant assignment of the eight cells. **(5 Marks)** **(c)** State two advantages and two limitations of min-cut placement. **(2 Marks)**	10	4	K3

Net	n1	n2	n3	n4	n5	n6	n7	n8	n9	n10
Cells	A, C	C, G	A, H	G, H	A, D	C, D	E, F	E, G	B, C	D, G

5		A two-layer channel has the pin assignment given below; 0 denotes no connection. The channel is to be routed by the left-edge algorithm. **(a)** Write the horizontal interval of each net, form the column sets S(i), and determine the channel density. What lower bound does the density place on the number of tracks? **(3 Marks)** **(b)** Construct the vertical constraint graph and verify that it is acyclic, so that a dogleg-free routing exists. **(2 Marks)** **(c)** Apply the left-edge algorithm. For every track, state the nets placed on it in the order they are selected, and the reason a net is skipped. Sketch the routed channel. **(4 Marks)** **(d)** State the number of tracks used and compare it with the density. Is the routing optimal in the number of tracks? **(1 Mark)**	10	5	K3

Column	1	2	3	4	5	6	7	8	9	10
TOP	1	0	3	0	4	2	0	4	6	5
BOTTOM	0	2	0	1	3	6	5	7	0	7

**********All the best **********